

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – AI ScamShield: Intelligent Scam & Phishing Detection Agent

Domain of Project – Cybersecurity

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Problem Statement -

Problem Statement: Ai Scamshield - Intelligent Scam & Phishing Detection Agentt

TheChallenge

Online scams and cyber fraud are increasing rapidly through phishing emails, fake banking messages, malicious websites, QR code scams, fake job offers, investment fraud, and WhatsApp scams. Many users cannot distinguish genuine messages from fraudulent ones, resulting in financial loss, identity theft, and compromised personal information. Existing solutions often require technical expertise and do not provide real-time personalized guidance.

The Objective

The objective of AI ScamShield is to build an AI-powered cybersecurity assistant using IBM watsonx.ai, IBM Granite Models, LangFlow, and Retrieval-Augmented Generation (RAG). The system analyzes suspicious messages, emails, URLs, and QR codes, identifies potential scams, explains the reasons behind the detection, retrieves trusted cybersecurity information, and provides personalized safety recommendations through a multi-agent AI architecture.

AI ScamShield uses a multi-agent architecture where each AI agent performs a specific task, such as scam detection, threat analysis, knowledge retrieval, and cyber safety guidance. Powered by IBM Granite Models and LangFlow, the system delivers accurate, context-aware responses and helps users identify online threats while promoting safe digital practices. It also uses RAG to retrieve trusted cybersecurity information, ensuring reliable and explainable recommendations.

As cyber threats continue to evolve, there is a growing need for intelligent systems that can detect scams before users become victims. AI ScamShield provides real-time scam detection, explains suspicious patterns in simple language, and offers preventive guidance using trusted cybersecurity knowledge. By combining IBM watsonx.ai, IBM Granite Models, LangFlow, and RAG, the solution improves digital awareness and helps users make safer online decisions with confidence.

Proposed Solution -

The proposed solution is an AI-powered ScamShield Agent built using IBM watsonx.ai, IBM Granite Models, and LangFlow to provide intelligent scam detection and cybersecurity guidance. The system analyzes suspicious emails, SMS messages, WhatsApp chats, banking alerts, URLs, and QR codes to identify potential scams, assess risk levels, and provide personalized security recommendations based on the content and threat type.

The solution includes four intelligent agents:

Scam Detection Agent – Detects phishing attempts, fake messages, and online scams while assigning a risk level.

Threat Analysis Agent – Examines suspicious patterns, malicious links, impersonation attempts, and social engineering techniques.

Knowledge Retrieval Agent – Retrieves trusted cybersecurity information using RAG from sources such as CERT-In, RBI, and OWASP.

Cyber Safety Advisor Agent – Provides preventive measures, reporting guidance, and personalized cybersecurity recommendations.

A user-friendly dashboard allows users to analyze suspicious content, view threat reports, and receive AI-generated cybersecurity guidance. IBM watsonx.ai and IBM Granite Models power the AI responses, while LangFlow manages the overall multi-agent workflow, delivering an intelligent, reliable, and seamless cybersecurity experience.

Technology Used

LangFlow Platform – For designing and orchestrating the multi-agent cybersecurity workflow visually.

IBM Granite Model (e.g., ibm-granite-3-3-8b-instruct) – For natural language understanding, scam detection, threat analysis, reasoning, and personalized cybersecurity recommendations.

IBM Cloud Lite – Provides a secure, scalable cloud platform for developing, deploying, and managing the AI-powered ScamShield Agent.

IBM watsonx.ai – For AI model deployment, prompt engineering, model inference, and AI governance.

Retrieval-Augmented Generation (RAG) – Retrieves trusted cybersecurity knowledge before generating accurate and context-aware scam analysis.

Vector Database (FAISS/ChromaDB) – Stores and retrieves cybersecurity resources such as phishing guidelines, scam patterns, CERT-In advisories, RBI security guidelines, and OWASP documentation for RAG.

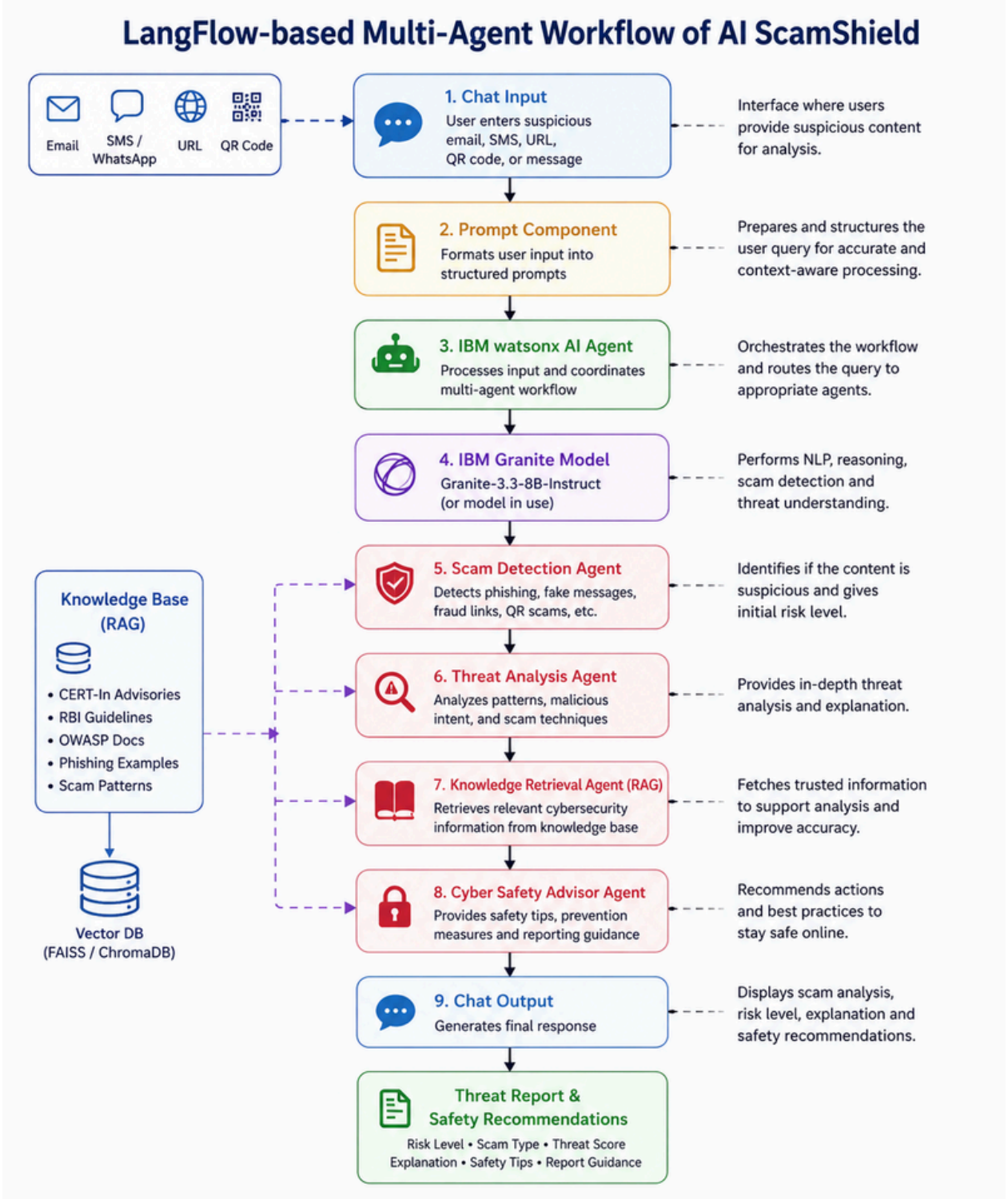
Python Flask – Backend framework used to build the AI ScamShield application and APIs.

Bootstrap 5, HTML, CSS, JavaScript – Used to create a responsive, interactive, and user-friendly cybersecurity dashboard.

Langflow component Used -

- 1) Chat Input – Interface where users enter suspicious emails, SMS messages, WhatsApp chats, banking messages, URLs, or QR code content for scam analysis.
- 2) IBM watsonx AI Agent – Processes user input using IBM Granite Models to detect scams, analyze threats, and generate personalized cybersecurity recommendations.
- 3) IBM Granite Model – Granite-3.3-8B-Instruct (or the Granite model you actually use) – Performs natural language understanding, scam detection, threat reasoning, and cybersecurity analysis.
- 4) Chat Output – Displays AI-generated scam analysis, risk level, scam category, confidence score, threat explanation, and cyber safety recommendations.
- 5) File Component – Stores and retrieves cybersecurity resources such as phishing examples, CERT-In advisories, RBI security guidelines, OWASP documentation, and scam knowledge for AI processing.
- 6) Prompt Component – Formats user queries and suspicious content into structured prompts before sending them to the IBM Granite Model for accurate, context-aware, and explainable scam detection and cybersecurity guidance.

Langflow workflow - Langflow-based Multi-Agent Workflow of the Ai - ScamShield



Architecture Blueprint of AI ScamShield

AI-Powered Multi-Agent System for Scam Detection and Cyber Safety

1. USER INPUT


Email


SMS


WhatsApp Message


URL / Link


QR Code


Other Text

2. FRONTEND

**Web Interface**
HTML, CSS, JavaScript, Bootstrap 5

- ✓ Submit suspicious content
- ✓ View analysis results
- ✓ Get safety recommendations

3. BACKEND

**Flask Backend**
Python Flask APIs

- ✓ Receives user input
- ✓ Handles requests
- ✓ Sends data to LangFlow workflow

4. LANGFLOW MULTI-AGENT WORKFLOW

**LangFlow Platform**
Orchestrates Multi-Agent Workflow

**1. Scam Detection Agent**
Identifies potential scams and assigns risk level

**2. Threat Analysis Agent**
Analyzes patterns, malicious indicators and intent

**3. Knowledge Retrieval Agent (RAG)**
Retrieves trusted cybersecurity knowledge and guidelines

**4. Cyber Safety Advisor Agent**
Generates safety tips, prevention measures and recommendations

5. AI MODEL & PLATFORM

**IBM watsonx.ai**
AI Model Deployment & Inference

**IBM Granite Model**
(e.g., ibm-granite-3-3-8b-instruct)
Natural Language Understanding, Reasoning & Generation

7. OUTPUT

**AI Scam Analysis Report**

 Risk Level

 Scam Type

 Threat Score

 Explanation

 Safety Recommendations

6. KNOWLEDGE BASE (RAG)

**Vector Database**
FAISS / ChromaDB

Cybersecurity Knowledge Sources

-  CERT-In Advisories
-  RBI Security Guidelines
-  OWASP Documentation
-  Common Scam Examples
-  Phishing & Malware Patterns
-  Cyber Safety Best Practices

8. KEY BENEFITS

- ✓ Real-time scam detection
- ✓ Explainable AI responses
- ✓ Trusted cybersecurity knowledge (RAG)
- ✓ Personalized safety recommendations
- ✓ User-friendly and secure platform

 Deployed on IBM Cloud Lite

 Secure

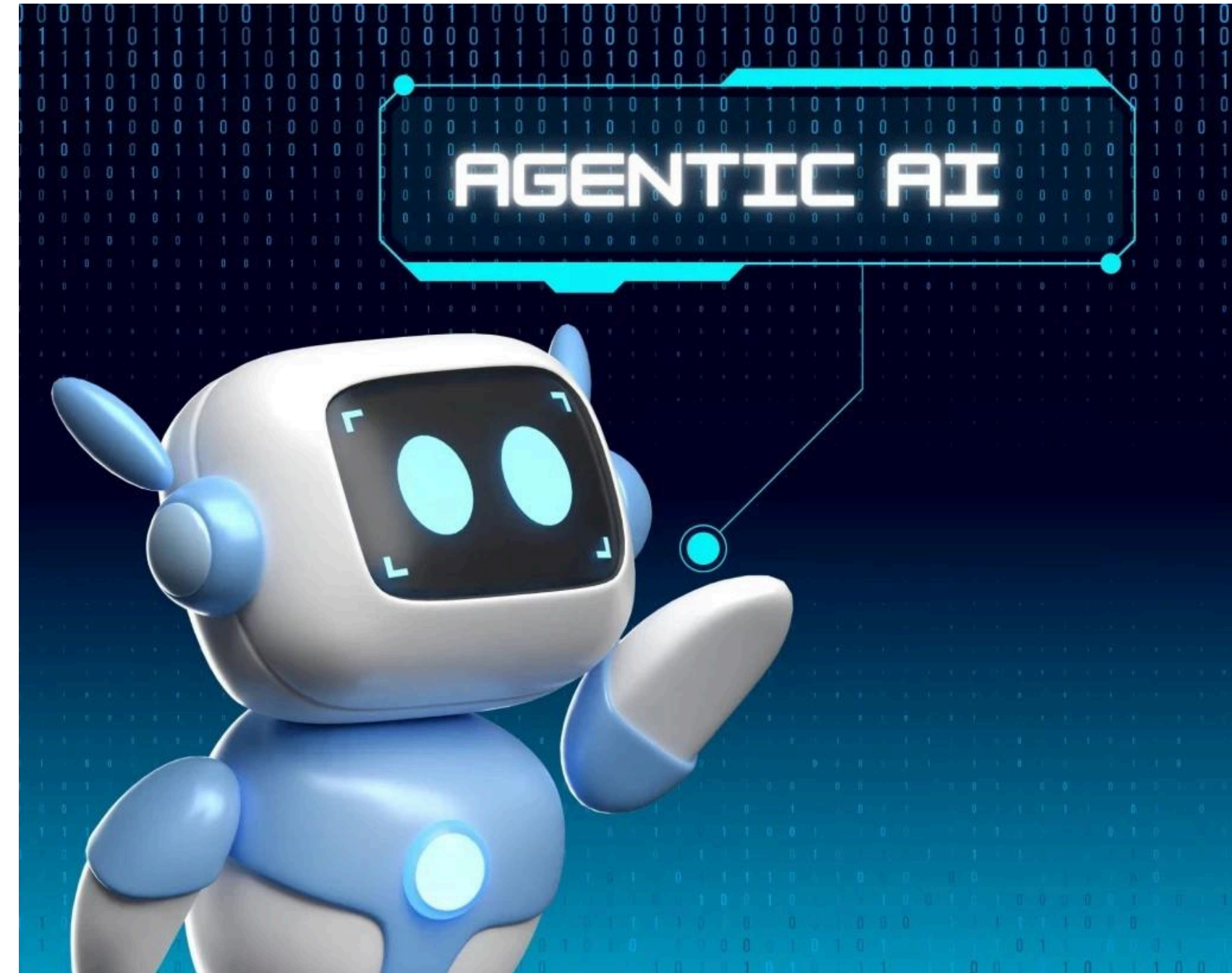
 Scalable

 Reliable

 Cost-Effective

Role of Agentic AI in the solution

Agentic AI is the core intelligence behind AI ScamShield, enabling multiple specialized AI agents to work together to detect and analyze online scams efficiently. Each agent performs a specific task, such as identifying phishing attempts, analyzing suspicious patterns, retrieving trusted cybersecurity knowledge through Retrieval-Augmented Generation (RAG), and generating personalized safety recommendations. This collaborative approach improves the accuracy, reliability, and explainability of scam detection while providing users with real-time cyber safety guidance. Powered by IBM watsonx.ai, IBM Granite Models, and LangFlow, AI ScamShield delivers intelligent, context-aware, and user-friendly protection against phishing, fraud, and other online cyber threats.



Novelty and Uniqueness

AI ScamShield stands out by combining Agentic AI, IBM Granite Models, LangFlow, and Retrieval-Augmented Generation (RAG) to provide intelligent and explainable scam detection. Unlike traditional spam filters that only classify messages, AI ScamShield analyzes the context, identifies different types of cyber scams, explains the reasons behind the detection, retrieves trusted cybersecurity information from official sources, and provides personalized safety recommendations. Its multi-agent architecture enables accurate threat analysis, real-time decision-making, and an interactive cybersecurity dashboard, making it a scalable, user-friendly, and innovative solution for protecting users against evolving online fraud and phishing attacks.

Git Hub Link

<https://github.com/ubbushaik8-rgb/AI-Scamshield>

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Future Scope

AI ScamShield can be further enhanced by integrating with popular platforms such as Gmail, WhatsApp, SMS services, and web browsers for real-time scam detection. Future versions can include voice scam and deepfake detection, image and QR code fraud analysis, and multilingual support to reach a wider audience. The system can also leverage live threat intelligence feeds, advanced machine learning models, and enterprise security dashboards to provide proactive cyber threat monitoring. These enhancements will make AI ScamShield a comprehensive, scalable, and enterprise-ready cybersecurity solution for individuals, organizations, and government agencies.

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Thank You!

Thank you for your time and interest.